SANCHO

Spinless Analogue to Nonequilibrium Correlated Hopping

User manual

Aadi Konidena, King's College London

October 2026

Abstract

SANCHO computes the current–voltage characteristics of a tight-binding chain with an on-site Hubbard interaction, coupled to two leads. It uses self-consistent non-equilibrium Green's functions (NEGF), with the interaction treated to zeroth, first (Hartree–Fock), second or third order in U. Spin is not resolved: the chain is paramagnetic and the current carries a factor of two for spin. Unlike QUICHOTTE, which evaluates the second order by FFT, SANCHO evaluates every order through factorised particle–hole and particle–particle *bubbles*, which is what makes the third order affordable. SANCHO lives in the `SANCHO/` folder of the QUICHOTTE repository but is a complete, independent code with its own build, tests and examples.

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1 Quick start

```
cd QUICHOTTE/SANCHO
make            # builds bin/sancho.x (needs gfortran + BLAS/LAPACK)
make test       # 6 regression tests, ~10 s; all must PASS

mkdir -p ~/runs/chain && cd ~/runs/chain
cp ~/QUICHOTTE/SANCHO/examples/chain_N20_order2/input.dat .   # edit as needed
OMP_NUM_THREADS=4 ~/QUICHOTTE/SANCHO/bin/sancho.x
```

Each run lives in its own directory, *outside* the repository, which holds only its `input.dat`.

platform	build command	needs
macOS	`make`	`brew install gcc openblas`
Linux	`make`	`gfortran`, `libopenblas-dev`
KCL CREATE	`make PLATFORM=create`	`source hpc/create/modules.sh`
ARCHER2	`make PLATFORM=archer2`	`source hpc/archer2/modules.sh`

2 Model

Units: energies in eV, voltages in V, temperatures in K, $\hbar = e = 1$ (factors of $\hbar$ are kept in the equations and in the source), and $k_B = 8.6173303 \times 10^{-5}$ eV/K.

2.1 Hamiltonian

A chain of N sites (`N_sites`) with one orbital per site:

$$H = \sum_i \varepsilon\, n_i + t \sum_i \left(c_i^\dagger c_{i+1} + \text{h.c.}\right) + \sum_i U_i\, n_{i\uparrow} n_{i\downarrow}, \tag{1}$$

with $\varepsilon =$ `E_CC`, $t =$ `t_hop`, and $U_i =$ `U_int`, or one value per site from `hubbard_file`. Spin enters only through the interaction, and the solution is assumed paramagnetic, so the Green's functions are $N \times N$ matrices for one spin species (`build_chain_hamiltonian` in `src/hamiltonian.f90`).

This chain is only one choice: nothing else in the code depends on it, and any other spin-independent single-particle Hamiltonian can be used instead, as described in the section *Using your own Hamiltonian*.

2.2 Leads

Wide-band leads: Γ_L couples to site 1, Γ_R to site N ($\Gamma_L =$ `GammaL`, $\Gamma_R =$ `GammaR`, default 1 eV each). The bias is applied to the left lead only:

$$f_L(\omega) = \frac{1}{e^{\beta(\omega-\mu+V)}+1}, \qquad f_R(\omega) = \frac{1}{e^{\beta(\omega-\mu)}+1}, \tag{2}$$

i.e. $\mu_L = \mu - V$, $\mu_R = \mu$. With $U \neq 0$ the asymmetric bias makes the I–V curve not exactly odd in V.

3 Non-equilibrium Green's functions

3.1 Energy grid

$$\omega_k = \omega_{\text{init}} + k\delta, \quad k = 1, \ldots, N_\omega, \qquad \omega_{\text{init}} = \epsilon_{\min} - \texttt{dw}, \qquad N_\omega = \left\lfloor \frac{\epsilon_{\max} + \texttt{up} - \omega_{\text{init}}}{\delta} \right\rfloor. \tag{3}$$

3.2 Dyson and Keldysh equations

$$G^r(\omega) = \Big[\hbar\omega - H + \tfrac{i}{2\hbar}(\Gamma_L + \Gamma_R) - \Sigma^r(\omega)\Big]^{-1}, \qquad G^a = (G^r)^\dagger, \tag{4}$$

$$G^<(\omega) = G^r(\omega)\Big[\tfrac{i}{\hbar}\big(f_L\Gamma_L + f_R\Gamma_R\big) + \Sigma^<(\omega)\Big]G^a(\omega), \tag{5}$$

$$G^>(\omega) = G^<(\omega) + G^r(\omega) - G^a(\omega). \tag{6}$$

3.3 Occupations and current

$$G^<_{ii}(t{=}0) = \frac{\delta}{2\pi}\sum_k G^<_{ii}(\omega_k), \qquad n_i = -i\hbar\, G^<_{ii}(t{=}0) \;\; \text{(per spin)}, \tag{7}$$

$$J = 2\cdot\frac{1}{\hbar}\frac{\delta}{2\pi}\sum_k \mathrm{Re}\,\mathrm{Tr}\Big\{i\Gamma_L\Big[\big(f_L - 1\big)G^<(\omega_k) - f_L G^>(\omega_k)\Big]\Big\}, \tag{8}$$

where the factor 2 is the spin degeneracy. The bracket equals $-[G^< + f_L(G^r - G^a)]$, so this is minus the Meir–Wingreen expression; with this sign $J > 0$ for $V > 0$.

4 Interaction self-energies

All self-energies are built from the input Green's functions G of the current SCF step (`src/self_energies.f90`, first order in `interacting_gf`). Grid sums carry the weight $p = \delta/2\pi$.

4.1 First order (Hartree), `order` ≥ 1

$$\Sigma^{(1)r}_{ij} = -i\hbar\, U_i\, G^<_{ii}(t{=}0)\,\delta_{ij} = U_i\, n_i\,\delta_{ij}. \tag{9}$$

In the paramagnetic chain the opposite-spin density equals n_i, and the Fock term (same spin) does not appear for an on-site $Un_\uparrow n_\downarrow$ interaction.

4.2 Bubbles

`compute_bubbles` first copies G with the energy index first (contiguous loops), then forms, for every pair (a, b):

Particle–hole bubbles at transfer $\nu_m = m\delta$, $m = -(N_\omega - 1), \ldots, N_\omega - 1$:

$$\begin{aligned}
\chi^{>}_{ab}(\nu_m) &= p \sum_k G^{>}_{ab}(\omega_k)\, G^{<}_{ba}(\omega_k - \nu_m), \\
\chi^{<}_{ab}(\nu_m) &= p \sum_k G^{<}_{ab}(\omega_k)\, G^{>}_{ba}(\omega_k - \nu_m), \\
\chi^{r}_{ab}(\nu_m) &= p \sum_k \big[G^{r}_{ab}(\omega_k) G^{<}_{ba}(\omega_k - \nu_m) + G^{<}_{ab}(\omega_k) G^{a}_{ba}(\omega_k - \nu_m)\big], \\
\chi^{a}_{ab}(\nu_m) &= p \sum_k \big[G^{a}_{ab}(\omega_k) G^{<}_{ba}(\omega_k - \nu_m) + G^{<}_{ab}(\omega_k) G^{r}_{ba}(\omega_k - \nu_m)\big].
\end{aligned} \tag{10}$$

Particle–particle bubbles (order 3 only) at total energy $\Omega_n = 2\omega_{\text{init}} + n\delta$, $n = 2, \ldots, 2N_\omega$:

$$\begin{aligned}
\phi^{>}_{ab}(\Omega_n) &= p \sum_k G^{>}_{ab}(\omega_k) G^{>}_{ab}(\Omega_n - \omega_k), \\
\phi^{<}_{ab}(\Omega_n) &= p \sum_k G^{<}_{ab}(\omega_k) G^{<}_{ab}(\Omega_n - \omega_k), \\
\phi^{r}_{ab}(\Omega_n) &= p \sum_k \big[G^{r}_{ab} G^{>}_{ab} + G^{<}_{ab} G^{r}_{ab}\big], \\
\phi^{a}_{ab}(\Omega_n) &= p \sum_k \big[G^{a}_{ab} G^{>}_{ab} + G^{<}_{ab} G^{a}_{ab}\big],
\end{aligned} \tag{11}$$

where in the last line the first factor is at ω_k and the second at $\Omega_n - \omega_k$. All sums run only over terms that stay on the grid.

4.3 Second order, `order` ≥ 2

With $c_2 = -(i\hbar)^2 = \hbar^2$ (`second_order_sigma`):

$$\Sigma^{(2)<}_{ij}(\omega) = c_2 U_i U_j\, p \sum_m G^{<}_{ij}(\omega - \nu_m)\, \chi^{<}_{ij}(\nu_m), \tag{12}$$

$$\Sigma^{(2)>}_{ij}(\omega) = c_2 U_i U_j\, p \sum_m G^{>}_{ij}(\omega - \nu_m)\, \chi^{>}_{ij}(\nu_m), \tag{13}$$

$$\Sigma^{(2)r}_{ij}(\omega) = c_2 U_i U_j\, p \sum_m \big[G^{r}_{ij}(\omega - \nu_m)\, \chi^{>}_{ij}(\nu_m) + G^{<}_{ij}(\omega - \nu_m)\, \chi^{r}_{ij}(\nu_m)\big]. \tag{14}$$

In time this is the familiar second-Born form $\Sigma^{<}_{ij}(t) = \hbar^2 U_i U_j\, G^{<}_{ij}(t) G^{<}_{ij}(t) G^{>}_{ji}(-t)$.

4.4 Third order, `order` $= 3$

The third-order diagrams (the corrected Appendix B derivation) factorise into bubble chains summed over an internal site k (`third_order_sigma`):

$$\begin{aligned}
S^{>}_{ij}(\nu) &= \textstyle\sum_k U_k \big[\chi^{r}_{ik}\chi^{>}_{kj} + \chi^{>}_{ik}\chi^{a}_{kj}\big](\nu), \\
S^{<}_{ij}(\nu) &= \textstyle\sum_k U_k \big[\chi^{r}_{ik}\chi^{<}_{kj} + \chi^{<}_{ik}\chi^{a}_{kj}\big](\nu), \\
S^{r}_{ij}(\nu) &= \textstyle\sum_k U_k\, \chi^{r}_{ik}\chi^{r}_{kj}(\nu),
\end{aligned} \tag{15}$$

and $P_{ij}^{>,<,r}(\Omega)$ the same with $\chi \to \phi$. Then, with $c_3 = -(i\hbar)^3$,

$$\Sigma_{ij}^{(3)<}(\omega) = c_3 U_i U_j \, p\Big[\sum_m G_{ij}^{<}(\omega-\nu_m) S_{ij}^{<}(\nu_m) + \sum_n G_{ji}^{>}(\Omega_n-\omega) P_{ij}^{<}(\Omega_n)\Big], \tag{16}$$

$$\Sigma_{ij}^{(3)>}(\omega) = c_3 U_i U_j \, p\Big[\sum_m G_{ij}^{>}(\omega-\nu_m) S_{ij}^{>}(\nu_m) + \sum_n G_{ji}^{<}(\Omega_n-\omega) P_{ij}^{>}(\Omega_n)\Big], \tag{17}$$

$$\Sigma_{ij}^{(3)r}(\omega) = c_3 U_i U_j \, p\Big[\sum_m \big(G_{ij}^{r} S_{ij}^{>} + G_{ij}^{<} S_{ij}^{r}\big) + \sum_n \big(G_{ji}^{a} P_{ij}^{>} + G_{ji}^{>} P_{ij}^{r}\big)\Big], \tag{18}$$

with the same arguments as in the first two lines. $\Sigma^{>}$ is not needed by the SCF; it is kept so that $\Sigma^r - \Sigma^a = \Sigma^{>} - \Sigma^{<}$ can be checked.

4.5 Total self-energy and cost

$\Sigma^r = \Sigma^{(1)} + \Sigma^{(2)r} + \Sigma^{(3)r}$ and $\Sigma^{<} = \Sigma^{(2)<} + \Sigma^{(3)<}$, up to `order`. The bubbles cost $\mathcal{O}(N^2 N_\omega^2)$ per SCF step; the third-order sums over k add $\mathcal{O}(N^3 N_\omega)$. Besides $8N^2N_\omega$ complex numbers for $G_{\rm in}$ and $G_{\rm out}$, the self-energy work arrays take $\approx 15\,N^2N_\omega$ complex numbers at order 2 and $\approx 38\,N^2N_\omega$ at order 3 (the third-order arrays are allocated only when `order = 3`).

5 Self-consistency

The SCF machinery is the same as in QUICHOTTE and lives in the shared files `common/broyden_mixing.f90` and `common/scf_checks.f90`:

- G^r and the independent half of $G^{<}$ (anti-Hermitian) on the whole grid are packed into a real vector x; the residual is $F(x) = x_{\rm out}(x) - x$.
- $F = 0$ is solved by Johnson's modified Broyden mixing (Phys. Rev. B **38**, 12807 (1988)), with weight `alpha_mix`, a ring buffer of `nhist` steps (0 = linear mixing) and regulariser $w_0 = 0.01$: $x_{n+1} = x_n + \alpha F_n - \sum_k \gamma_k u_k$, $\gamma = (w_0^2 \mathbb{1} + \Delta F^T \Delta F)^{-1} \Delta F^T F_n$.
- Converged when $\max|F| <$ `tolf`, tested before mixing; at most `maxscf` steps. The history is in `scf_V_<V>.dat`.
- Sanity checks after every step, for every site: occupation $n_i = -i\hbar\, p \sum_k G_{ii}^{<}$ and spectral weight $W_i = -\hbar\,(\delta/\pi) \sum_k \mathrm{Im}\, G_{ii}^{r}$. NaN or $n_i \notin [-1, 2]$ abandons the voltage; $n_i \notin [-0.05, 1.05]$ or $W_i < 0.5$ rejects a converged solution.
- The first voltage starts from the non-interacting G_0. With `continuation = T` (the default) every later voltage starts from the previous converged one; a failed voltage is flagged (column $3 = 0$), and the next one starts from G_0. Exit code 3 if any voltage failed.

Continuation pays off for small `delv`. For coarse voltage steps it can start too far from the new solution (the Broyden iteration may then oscillate); use `continuation = F` in that case.

6 Input reference: `input.dat`

One `key = value` per line, any order, `#` starts a comment. Missing required keys and duplicated keys stop the run with a full list; unknown keys give a warning. Old names in the third column are still accepted.

key	default	old name	meaning
V, Vf, delv	required		first voltage, last voltage, step [V]
T	required		temperature [K]
mu	required		chemical potential at $V = 0$ [eV]
hamiltonian	chain		model: chain or custom (see Using your own Hamiltonian)
N_sites	required	Nsites, Natoms	number of sites N
E_CC	required		on-site energy [eV]
t_hop	required		hopping [eV]
U_int	required		uniform U [eV]
hubbard_file	none		file with one U per site (overrides U_int)
order	required		0, 1, 2 or 3
GammaL, GammaR	1.0		lead couplings to sites 1 and N [eV]
dw, up	required		grid margins below/above the spectrum [eV]
delta	required		grid spacing [eV]
alpha_mix	0.3		Broyden linear-mixing weight
nhist	8		Broyden history length (0 = linear mixing)
maxscf	200		maximum SCF steps per voltage
tolf	10^{-6}		convergence threshold on $\max \lvert F \rvert$
continuation	T		start each voltage from the previous one
restart, ver	–		accepted for old files, not used

7 Output reference

file	content
Volt_Current_<order>.dat	V, J, converged (1/0)
populations.dat	one line per site per voltage: V, site, n_i (per spin), $2n_i$
spectral_function.dat	V, ω, $A_{ii}(\omega) = -2 \operatorname{Im} G^r_{ii}$ for every site; one block per voltage
scf_V_<V>.dat	SCF history: step, $\max \lvert F \rvert$, $\mathrm{rms} \lvert F \rvert$, history length used
info_Hamiltonian.dat	H, Hermiticity check, eigenvalues, eigenvectors
runtime_datasheet.dat	code version (git describe), voltages processed, wall time

Exit codes: 0 all voltages converged, 1 bad input, 3 some voltages failed.

8 Workflow

8.1 Single runs on a cluster

hpc/create/job.slurm, hpc/young/job.sge, hpc/archer2/job.slurm run SANCHO in the directory they are submitted from (set EXE once at the top).

8.2 Parameter scans on ARCHER2

hpc/archer2/sweep/ runs many cases as one Slurm job array, packing several runs onto each 128-core node. The runs are listed in manifest.txt: the columns are input keys, and comma-separated values expand to every combination. The extra column Ut sets $U = Ut \cdot |t|$.

```
./generate.sh        # runs/<name>/input.dat for every manifest line
./submit.sh          # submit;  ./submit.sh failed  resubmits failures
./status.sh          # state of every run
./collect.sh         # all I-V curves -> results_IV.dat
```

See `hpc/archer2/sweep/README.md`. These scripts (and the job scripts) are the only shell code; SANCHO itself and its tools are Fortran.

8.3 Tools

`make tools`: `bin/dIdV.x [file] [column]` (dJ/dV to `dI_dV.dat`) and the experimental `bin/stm_junction.x` (STM tip + surface geometry for the unfinished `build_hamiltonian_from_xyz`).

9 Using your own Hamiltonian

The nearest-neighbour chain of the Model section is only one model. Nothing else in SANCHO depends on it: the Green's functions, bubbles, self-energies, SCF loop and observables see nothing but the matrix H. The input key `hamiltonian` chooses which model builds H:

`hamiltonian =`	routine in `src/hamiltonian.f90`
`chain (default)`	`build_chain_hamiltonian`
`custom`	`build_custom_hamiltonian`

The number of sites is always the key `N_sites`. The dispatcher `build_hamiltonian` calls the chosen routine, so the main program never needs editing.

9.1 Your own model in three steps

As shipped, `build_custom_hamiltonian`, the last routine in `src/hamiltonian.f90`, is the plain chain written out explicitly, so `hamiltonian = custom` runs straight away:

```
subroutine build_custom_hamiltonian()
   integer :: i

   H = (0.d0, 0.d0)
   do i = 1, Nsites
      H(i, i) = E_CC                        !...on-site energy
   end do
   do i = 1, Nsites - 1
      H(i, i + 1) = t_hop                   !...bond i -> i+1
      H(i + 1, i) = conjg(H(i, i + 1))      !...keep H Hermitian
   end do
end subroutine build_custom_hamiltonian
```

(`Nsites` is the variable that holds the input key `N_sites`.)

1. Replace the body of `build_custom_hamiltonian` with your model. All run parameters (`Nsites`, `E_CC`, `t_hop`, ...) are available inside it. New parameters: declare them in

`src/parameters.f90` and read them in `read_input` (`src/input_output.f90`) with `call input_get('my_key', my_var)`; add `default=...` to make a key optional.

2. In `input.dat` set `hamiltonian = custom`.

3. Build (`make`) and check one run. `info_Hamiltonian.dat` should show your H, all-zero Hermiticity differences and the expected eigenvalues. With `order = 0`, the current at $V = 0$ should be zero. Then run `make test`, and add a small test case for your model (see Testing and validation) so that later changes cannot break it unnoticed.

9.2 What the matrix must look like

1. **Storage.** H is the module variable `H` of `src/hamiltonian.f90`, declared `complex(dp), allocatable :: H(:,:)`. The main program allocates it as `H(Nsites, Nsites)` before your routine is called; your routine only fills it (do not reallocate it).

2. **Basis.** One orbital per site, index i = site i, with $i = 1, \ldots, N_{\text{sites}}$.

3. **Spin-independent.** SANCHO assumes the same H for both spins (a paramagnetic chain; the current carries a factor 2). Spin-dependent terms (spin–orbit coupling, magnetic fields) cannot be expressed here; use QUICHOTTE for those.

4. **Hermitian.** $H = H^\dagger$, with a real diagonal. SANCHO has no `make_hermitian`, so fill both triangles, e.g. by setting `H(j,i) = conjg(H(i,j))` after each off-diagonal element, as the template does. `info_Hamiltonian.dat` lists $H_{ij} - H^*_{ji}$ for every pair; all must be zero.

5. **Units and energy zero.** Energies in eV, on the same scale as `mu`: at $V = 0$ the levels below μ are filled.

6. **Static and non-interacting.** H must not depend on energy or on the SCF iteration. The interaction enters only through the Hubbard U of each site, so do *not* put Hubbard or Hartree terms into H: they would be counted twice.

7. **Contacts.** The left lead couples to site 1 and the right lead to site N_{sites} (strengths `GammaL`, `GammaR`). Number your sites so that the contacted ones come first and last, or change the two lines that set `GammaL` and `GammaR` in `src/sancho.f90`.

Anything else is allowed. Hoppings may connect any pair of sites and have any range, on-site energies may differ from site to site, and the matrix may be dense. The bubbles and the second- and third-order self-energies are computed for every pair of sites, so they do not assume any particular connectivity. The energy grid needs no change: it is built from the eigenvalues of whatever H you supply.

9.3 Several models of your own

The `custom` slot is the quickest way to try a model. To keep several models side by side, give each one its own routine in `src/hamiltonian.f90` and register it with a single line in `build_hamiltonian`:

```
case ('ssh');  call build_ssh_hamiltonian()
```

Then select it in `input.dat` with `hamiltonian = ssh`. Nothing else needs to change: `build_hamiltonian` stops with an error if it is given a name it does not know. Parameters that the new model needs are read in `read_input`, as in step 1 above.

10 Code structure

file	module	content
src/sancho.f90	(program)	set-up, grid, voltage loop, output
src/parameters.f90	Parameters	run parameters, sizes, unit numbers
src/hamiltonian.f90	Hamiltonian	chain H; experimental H from an .xyz file
src/green_functions.f90	GreenFunctions	storage of G, Γ, grid; G_0; $G^<(t{=}0)$
src/self_energies.f90	SelfEnergies	bubbles, $\Sigma^{(2)}$, $\Sigma^{(3)}$
src/scf.f90	SCF	SCF driver, residual, Dyson/Keldysh (incl. $\Sigma^{(1)}$)
src/observables.f90	Observables	current, populations, spectral function
src/input_output.f90	InputOutput	reads input.dat, per-site U, printout
common/*.f90	(shared)	constants, linear algebra, input parser, Broyden, SCF checks

The files in common/ are identical to QUICHOTTE's; from the repository root, make check-common verifies this (CI runs it on every push). G_in is the input of the current SCF step and G_out its output; components %r, %a, %L, %G, each (Nsites, Nsites, n_omega).

11 Testing and validation

make test builds the Fortran runner tests/run_tests.f90 and runs the cases in tests/cases.txt, comparing each output number by number with tests/cases/<case>/reference/ ($|x - x_{\mathrm{ref}}| \leq 10^{-6} + 10^{-5}|x_{\mathrm{ref}}|$; override with TOL_ABS, TOL_REL).

case	what it covers
order0_noninteracting	4-site chain, $U = 0$
order1_hartree_fock	first order
order2_second_order	bubbles, second order, Broyden
order3_third_order	third order
order2_asymmetric_leads	$\Gamma_L \neq \Gamma_R$
legacy_natoms	old key names, no Broyden keys (defaults)

After a deliberate physics change: check the numbers, then make update-references, and commit the references with the change.

Validation history. The restructured code reproduces the previous version (SpinlessJunction) to better than 3×10^{-8} in the currents for orders 1–3 (October 2026). Before the clean-up, according to its commit history, the code agreed with four other independent codes.

12 Known limitations

- Perturbation theory up to third order in U; paramagnetic solutions only.
- Near half filling at larger $U/|t|$ the Broyden iteration can oscillate without converging. For example, a 20-site chain at $U/|t| = 2$, order 2, $V = 0$ does not converge within 200 steps with alpha_mix = 0.3, in the original code as well as in this one. Lower alpha_mix, raise maxscf, or approach the point in small voltage steps with continuation = T.

- Wide-band leads on the two end sites; bias on the left lead only.
- Uniform energy grid; voltages swept upwards only.
- The xyz Hamiltonian (`build_hamiltonian_from_xyz`) is a placeholder: element table and bond cutoff are not physical yet.